Geometric representation of complex numbers — Solutions

Warm-up

Question 1a:
To convert $z = 8 + 3i$ to modulus-argument form, we find the modulus r and the argument θ.

$$\begin{aligned} r = |z| &= \sqrt{8^2 + 3^2} \\ &= \sqrt{64 + 9} \\ &= \sqrt{73} \\ \theta = \mathrm{Arg}(z) &= \tan^{-1}\left(\frac{3}{8}\right) \approx 20.56^\circ \text{ or } 0.359 \text{ rad} \end{aligned}$$

Thus, the modulus-argument form is $\sqrt{73}(\cos(20.56^\circ) + i\sin(20.56^\circ))$.

Question 1b:
For $z = 2 - 6i$, the point lies in the fourth quadrant.

$$\begin{aligned} r &= \sqrt{2^2 + (-6)^2} \\ &= \sqrt{4 + 36} \\ &= \sqrt{40} = 2\sqrt{10} \\ \theta &= \tan^{-1}\left(\frac{-6}{2}\right) = \tan^{-1}(-3) \approx -71.57^\circ \text{ or } -1.249 \text{ rad} \end{aligned}$$

Thus, the modulus-argument form is $2\sqrt{10}(\cos(-71.57^\circ) + i\sin(-71.57^\circ))$.

Question 1c:
For $z = -1 - 2i$, the point lies in the third quadrant.

$$\begin{aligned} r &= \sqrt{(-1)^2 + (-2)^2} \\ &= \sqrt{1 + 4} \\ &= \sqrt{5} \\ \theta &= \tan^{-1}\left(\frac{-2}{-1}\right) - \pi = \tan^{-1}(2) - \pi \approx 63.43^\circ - 180^\circ = -116.57^\circ \text{ or } -2.034 \text{ rad} \end{aligned}$$

Thus, the modulus-argument form is $\sqrt{5}(\cos(-116.57^\circ) + i\sin(-116.57^\circ))$.

Question 1d:
For $z = -8 + 7i$, the point lies in the second quadrant.

$$\begin{aligned} r &= \sqrt{(-8)^2 + 7^2} \\ &= \sqrt{64 + 49} \\ &= \sqrt{113} \\ \theta &= \tan^{-1}\left(\frac{7}{-8}\right) + \pi = \tan^{-1}(-0.875) + \pi \approx -41.19^\circ + 180^\circ = 138.81^\circ \text{ or } 2.423 \text{ rad} \end{aligned}$$

Thus, the modulus-argument form is $\sqrt{113}(\cos(138.81^\circ) + i\sin(138.81^\circ))$.

Question 2a:
We evaluate the trigonometric functions for $\theta = \frac{\pi}{4}$.

$$\begin{aligned} z &= 2\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right) \\ &= 2\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) \\ &= \sqrt{2} + i\sqrt{2} \end{aligned}$$

Question 2b:
We evaluate the trigonometric functions for $\theta = -\frac{\pi}{6}$.

$$\begin{aligned} z &= \sqrt{3}\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right) \\ &= \sqrt{3}\left(\frac{\sqrt{3}}{2} + i\left(-\frac{1}{2}\right)\right) \\ &= \frac{3}{2} - \frac{\sqrt{3}}{2}i \end{aligned}$$

Question 2c:
For $\theta = \frac{5\pi}{8}$, we use the calculator or half-angle formulas. $\cos(\frac{5\pi}{8}) \approx -0.3827$ and $\sin(\frac{5\pi}{8}) \approx 0.9239$.

$$\begin{aligned} z &= \sqrt{2}\left(\cos\left(\frac{5\pi}{8}\right) + i\sin\left(\frac{5\pi}{8}\right)\right) \\ &= \sqrt{2}(-0.3827 + 0.9239i) \\ &\approx -0.541 + 1.307i \end{aligned}$$

Question 2d:
We evaluate the trigonometric functions for $\theta = \frac{3\pi}{4}$.

$$\begin{aligned} z &= \sqrt{5}\left(\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right) \\ &= \sqrt{5}\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) \\ &= -\frac{\sqrt{10}}{2} + i\frac{\sqrt{10}}{2} \end{aligned}$$

Question 3a:
Using the property $|z \times w| = |z| \times |w|$.

$$\begin{aligned}|z \times w| &= 3 \times 7 \\ &= 21\end{aligned}$$

Question 3b:
Using the property $|z \times w| = |z| \times |w|$.

$$\begin{aligned}|z \times w| &= 1 \times 4 \\ &= 4\end{aligned}$$

Question 3c:
Using the property $|z \times w| = |z| \times |w|$.

$$\begin{aligned}|z \times w| &= 2 \times 6 \\ &= 12\end{aligned}$$

Question 3d:
Using the property $|z \times w| = |z| \times |w|$.

$$\begin{aligned}|z \times w| &= \sqrt{18} \times \sqrt{2} \\ &= \sqrt{36} \\ &= 6\end{aligned}$$

Question 4a:
Using the property $\arg(z \times w) = \arg(z) + \arg(w)$.

$$\begin{aligned}\arg(z \times w) &= \frac{\pi}{3} + \frac{\pi}{3} \\ &= \frac{2\pi}{3}\end{aligned}$$

Question 4b:
Using the property $\arg(z \times w) = \arg(z) + \arg(w)$.

$\arg(\text{z} \times w) = \frac{\pi}{2} + \frac{\pi}{6}$
$= \frac{3\pi+\pi}{6}$
$= \frac{4\pi}{6} = \frac{2\pi}{3}$

Question 4c:
Using the property $\arg(z \times w) = \arg(z) + \arg(w)$.

$$
\begin{aligned}
\arg(z \times w) &= \frac{\pi}{5} + \frac{2\pi}{5} \\
&= \frac{3\pi}{5}
\end{aligned}
$$

Question 4d:
Using the property $\arg(z \times w) = \arg(z) + \arg(w)$.

$$
\begin{aligned}
\arg(z \times w) = \frac{\pi}{3} + \frac{\pi}{4} \\
&= \frac{4\pi + 3\pi}{12} \\
&= \frac{7\pi}{12}
\end{aligned}
$$

Question 5a:
Let $z = r(\cos\theta + i\sin\theta)$. We want $w \times z$ to be 3 times longer than z and rotated $30°$ anticlockwise.

$$
\begin{aligned}
|w \times z| &= 3|z| \\
|w| \times |z| &= 3|z| \implies |w| = 3 \\
\arg(w \times z) &= \arg(z) + 30° \\
\arg(w) + \arg(z) &= \arg(z) + 30° \implies \arg(w) = 30° = \frac{\pi}{6}
\end{aligned}
$$

Thus, $w = 3(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$.

Question 5b:
We want $w \times z$ to be the same length as z and rotated $90°$ anticlockwise.

$$
\begin{aligned}
|w \times z| &= |z| \implies |w| = 1 \\
\arg(w \times z) &= \arg(z) + 90° \implies \arg(w) = 90° = \frac{\pi}{2}
\end{aligned}
$$

Thus, $w = 1(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$.

Question 5c:
We want $w \times z$ to be half as long as z and rotated $45°$ clockwise.

$$
\begin{aligned}
|w \times z| &= \frac{1}{2}|z| \implies |w| = \frac{1}{2} \\
\arg(w \times z) &= \arg(z) - 45° \implies \arg(w) = -45° = -\frac{\pi}{4}
\end{aligned}
$$

Thus, $w = \frac{1}{2}(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4}))$.

Question 5d:
We want $w \times z$ to be 4 times longer than z and pointing in the opposite direction.

$$|w \times z| = 4|z| \implies |w| = 4$$
$$\arg(w \times z) = \arg(z) + 180^\circ \implies \arg(w) = 180^\circ = \pi$$

Thus, $w = 4(\cos \pi + i \sin \pi)$.

Skill-building

Question 1a:
Given $|z| = \sqrt{2}$ and $\text{Arg}(z) = \frac{\pi}{4}$.

$$|z^2| = |z|^2 = (\sqrt{2})^2 = 2$$
$$\text{Arg}(z^2) = 2\text{Arg}(z) = 2\left(\frac{\pi}{4}\right) = \frac{\pi}{2}$$

Question 1b:
A complex number is purely imaginary if its argument is $\frac{\pi}{2} + k\pi$ for $k \in \mathbb{Z}$.

$$\begin{aligned} \text{Arg}(z^n) &= n\text{Arg}(z) = n\frac{\pi}{4} \\ n\frac{\pi}{4} &= \frac{\pi}{2} + k\pi \\ n &= 2 + 4k \end{aligned}$$

Thus, z^n is purely imaginary for $n = 2, 6, 10, \ldots$ (or $n = 4k + 2$ for $k \in \mathbb{Z}$).

Question 1c:
A complex number is purely real if its argument is $k\pi$ for $k \in \mathbb{Z}$.

$$\begin{aligned} \text{Arg}(z^n) &= n\frac{\pi}{4} \\ n\frac{\pi}{4} &= k\pi \\ n &= 4k \end{aligned}$$

Thus, z^n is purely real for $n = 4, 8, 12, \ldots$ (or $n = 4k$ for $k \in \mathbb{Z}$).

Question 1d:
The first 4 powers of z are:

$$\begin{aligned} z^1 &= \sqrt{2}\text{cis}(\pi/4) \\ z^2 &= 2\text{cis}(\pi/2) \\ z^3 &= 2\sqrt{2}\text{cis}(3\pi/4) \\ z^4 &= 4\text{cis}(\pi) \end{aligned}$$

On an Argand diagram, these points spiral outwards from the origin, each rotated by 45° and with the modulus increasing by a factor of $\sqrt{2}$ each time.

Question 2:
Let $z_A = 1 + 3i$. The transformation is a rotation of 45° clockwise and an enlargement by $\sqrt{10}$.

$$\begin{aligned}
w &= z_A \times \sqrt{10}\text{cis}(-45^\circ) \\
w &= (1+3i) \times \sqrt{10}\left(\cos(-45^\circ) + i\sin(-45^\circ)\right) \\
w &= (1+3i) \times \sqrt{10}\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) \\
w &= (1+3i) \times \frac{\sqrt{20}}{2}(1-i) \\
w &= (1+3i) \times \sqrt{5}(1-i) \\
w &= \sqrt{5}(1 - i + 3i - 3i^2) \\
w &= \sqrt{5}(1 + 2i + 3) \\
w &= \sqrt{5}(4 + 2i) \\
w &= 4\sqrt{5} + 2\sqrt{5}i
\end{aligned}$$

Question 3:
Let $z_A = -4 - 3i$ and z_Z be the complex number for point Z. The condition $\angle OZA = 90^\circ$ and $|ZA| = 2|OZ|$ implies that the vector $\vec{ZA}$ is a 90° rotation of $\vec{OZ}$ scaled by 2.

$$\begin{aligned}
z_A - z_Z &= 2i(z_Z - 0) \quad \text{or} \quad z_A - z_Z &= -2i(z_Z - 0) \\
z_A - z_Z &= 2iz_Z
\end{aligned}$$

Solving for z_Z:

$$\begin{aligned}
z_A &= z_Z(1+2i) \\
z_Z &= \frac{z_A}{1+2i} \\
z_Z &= \frac{-4-3i}{1+2i} \\
z_Z &= \frac{(-4-3i)(1-2i)}{(1+2i)(1-2i)} \\
z_Z &= \frac{-4+8i-3i+6i^2}{1^2+2^2} \\
z_Z &= \frac{-4+5i-6}{5} \\
z_Z &= \frac{-10+5i}{5} \\
z_Z &= -2+i
\end{aligned}$$

Question 4:
This is a geometric interpretation problem.

a) $z+i$: Shift z up by 1 unit on the imaginary axis.

b) $z-(1+2i)$: Shift z left by 1 unit and down by 2 units.

c) iz: Rotate z by 90° anticlockwise about the origin.

d) z^2: Square the modulus and double the argument.

e) z^3: Cube the modulus and triple the argument.

f) $\frac{1}{z}$: Invert the modulus and negate the argument (reflection across real axis and scaling).

g) $-\bar{z}$: Reflect z across the real axis, then reflect across the origin (equivalent to reflection across the imaginary axis).

h) $z \times (\cos(-\frac{\pi}{6}) + i\sin(-\frac{\pi}{6}))$: Rotate z by 30° clockwise about the origin.

i) $\sqrt{z}$: The modulus is $\sqrt{|z|}$ and the arguments are $\frac{\text{Arg}(z)}{2}$ and $\frac{\text{Arg}(z)}{2}+\pi$.

Easier Exam Questions

Question 1:
We need the modulus and argument of $z=-1-i$.

$$|z| = \sqrt{(-1)^2+(-1)^2} = \sqrt{2}$$
$$\text{Arg}(z) = \tan^{-1}\left(\frac{-1}{-1}\right) - \pi = \tan^{-1}(1) - \pi = \frac{\pi}{4} - \pi = -\frac{3\pi}{4}$$

Comparing with the options, the correct answer is (B).

Question 2a:
For $z_1 = -\sqrt{2}+\sqrt{2}i$:

$$|z_1| = \sqrt{(-\sqrt{2})^2+(\sqrt{2})^2} = \sqrt{2+2} = 2$$
$$\text{Arg}(z_1) = \tan^{-1}\left(\frac{\sqrt{2}}{-\sqrt{2}}\right) + \pi = \tan^{-1}(-1) + \pi = -\frac{\pi}{4} + \pi = \frac{3\pi}{4}$$

Thus, $z_1 = 2\text{cis}(\frac{3\pi}{4})$.
For $z_2 = \sqrt{3}+i$:

$$|z_2| = \sqrt{(\sqrt{3})^2+1^2} = \sqrt{3+1} = 2$$
$$\text{Arg}(z_2) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

Thus, $z_2 = 2\text{cis}(\frac{\pi}{6})$.

Question 2b:
The vectors $O\vec{Z}_1$ and $O\vec{Z}_2$ are plotted on the Argand diagram starting from the origin to the points $(-\sqrt{2}, \sqrt{2})$ and $(\sqrt{3}, 1)$ respectively.

Question 2c:
Using the properties of arguments:

$$\begin{aligned}\arg\left(\frac{z_1}{z_2}\right) &= \arg(z_1) - \arg(z_2) \\ &= \frac{3\pi}{4} - \frac{\pi}{6} \\ &= \frac{9\pi - 2\pi}{12} = \frac{7\pi}{12}\end{aligned}$$

For $\arg(z_1 + z_2)$:

$$\begin{aligned}z_1 + z_2 &= (-\sqrt{2} + \sqrt{3}) + i(\sqrt{2} + 1) \\ \arg(z_1 + z_2) &= \tan^{-1}\left(\frac{\sqrt{2}+1}{\sqrt{3}-\sqrt{2}}\right) \approx 1.78 \text{ rad}\end{aligned}$$

Question 3:
Given $\arg(z) = \frac{\pi}{5}$.

$$\begin{aligned}\arg(z^7) &= 7 \times \arg(z) \\ &= 7 \times \frac{\pi}{5} = \frac{7\pi}{5} \\ &= \frac{7\pi}{5} - 2\pi = \frac{7\pi - 10\pi}{5} = -\frac{3\pi}{5}\end{aligned}$$

Comparing with the options, the correct answer is (B).

Question 4:
Given $w = 2 - i(2\sqrt{3})$ and $\text{Arg}(zw) = \frac{\pi}{3}$.

$$\begin{aligned}|w| &= \sqrt{2^2 + (-2\sqrt{3})^2} = \sqrt{4+12} = 4 \\ \text{Arg}(w) &= \tan^{-1}\left(\frac{-2\sqrt{3}}{2}\right) = \tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}\end{aligned}$$

Using the property $\text{Arg}(zw) = \text{Arg}(z) + \text{Arg}(w)$:

$$\begin{aligned}\frac{\pi}{3} &= \text{Arg}(z) + \left(-\frac{\pi}{3}\right) \\ \text{Arg}(z) &= \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}\end{aligned}$$

Question 5:
We first simplify $z = \frac{5-i}{2-3i}$ by multiplying by the conjugate.

$$\begin{aligned} z &= \frac{(5-i)(2+3i)}{(2-3i)(2+3i)} \\ &= \frac{10+15i-2i-3i^2}{2^2+3^2} \\ &= \frac{10+13i+3}{13} \\ &= \frac{13+13i}{13} = 1+i \end{aligned}$$

Now find the modulus and argument:

$$\begin{aligned} |z| &= \sqrt{1^2+1^2} = \sqrt{2} \\ \text{Arg}(z) &= \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4} \end{aligned}$$

Question 6:
We need the modulus and argument of $z = i - 1 = -1 + i$.

$$\begin{aligned} |z| &= \sqrt{(-1)^2+1^2} = \sqrt{2} \\ \text{Arg}(z) &= \tan^{-1}\left(\frac{1}{-1}\right) + \pi = \tan^{-1}(-1) + \pi = -\frac{\pi}{4} + \pi = \frac{3\pi}{4} \end{aligned}$$

Comparing with the options, the correct answer is (B).

Question 7a:
For $z = -\sqrt{3} + i$:

$$\begin{aligned} |z| &= \sqrt{(-\sqrt{3})^2+1^2} = \sqrt{3+1} = 2 \\ \text{Arg}(z) &= \tan^{-1}\left(\frac{1}{-\sqrt{3}}\right) + \pi = -\frac{\pi}{6} + \pi = \frac{5\pi}{6} \end{aligned}$$

Thus, $z = 2(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$.

Question 7b:
We want the smallest positive integer n such that $z^n \in \mathbb{R}$. This means $\text{Arg}(z^n) = k\pi$.

$$\begin{aligned} \text{Arg}(z^n) &= n \times \frac{5\pi}{6} = k\pi \\ n \times \frac{5}{6} &= k \\ n &= \frac{6k}{5} \end{aligned}$$

For n to be an integer, k must be a multiple of 5. The smallest positive $k = 5$ gives $n = 6$.

Harder Exam Questions

Question 1:
Let $z = x + iy$ with $y > 0$. Then $\bar{z} = x - iy$.

$$z\bar{z} = |z|^2 = x^2 + y^2$$
$$z - \bar{z} = (x + iy) - (x - iy) = 2iy$$

Now consider the expression $\frac{z\bar{z}}{z-\bar{z}}$:

$$\begin{aligned}\frac{z\bar{z}}{z-\bar{z}} &= \frac{x^2+y^2}{2iy} \\ &= \frac{x^2+y^2}{2iy} \times \frac{i}{i} \\ &= \frac{i(x^2+y^2)}{-2y} \\ &= -i\left(\frac{x^2+y^2}{2y}\right)\end{aligned}$$

Since $y > 0$ and $x^2 + y^2 > 0$, the term $\frac{x^2+y^2}{2y}$ is a positive real number. Thus, the complex number is purely imaginary with a negative imaginary part.

$$\arg\left(-i\frac{x^2+y^2}{2y}\right) = -\frac{\pi}{2}$$

Comparing with the options, the correct answer is (A).

Question 2:
Let $z_A = 2 + 3i$. Since $OACB$ is a square, the vector $\vec{OB}$ is a $90°$ rotation of $\vec{OA}$.

$$z_B = iz_A = i(2 + 3i) = 2i + 3i^2 = -3 + 2i$$

If the diagonal is $\vec{AC}$, then A and C are opposite vertices. Then O and B are opposite vertices. Then $\vec{OB} = \vec{OA} + \vec{OC}$.

$$z_C = z_B - z_A = (-3 + 2i) - (2 + 3i) = -5 - i$$

This matches option (A).

Question 3:
The angle θ between two vectors $\vec{V_1}$ and $\vec{V_2}$ is given by $\arg(\frac{V_2}{V_1})$.

$$\vec{AC} = c - a$$
$$\vec{BD} = d - b$$

Thus, the angle θ is:

$$\theta = \arg\left(\frac{c-a}{d-b}\right)$$

Comparing with the options, the correct answer is (C).

Question 4:
Based on the provided diagram, z is in the second quadrant. $\bar{z}$ is the reflection of z across the real axis, so $\bar{z}$ will be in the third quadrant. $i\bar{z}$ is a $90°$ anticlockwise rotation of $\bar{z}$. Since $\bar{z}$ is in the third quadrant, $i\bar{z}$ will be in the fourth quadrant.

Question 5a:
The vector $\vec{AB}$ is given by the difference of the complex numbers representing points B and A.

$$\begin{aligned}\vec{AB} &= z_B - z_A \\ &= (-2+4i) - (-6+2i) \\ &= -2+6+4i-2i \\ &= 4+2i\end{aligned}$$

Question 5b:
To find C, we rotate $\vec{AB}$ by $30°$ (since $\angle BAC = 30°$) and scale it by 2 (since $AC = 2AB$).

$$\begin{aligned}\vec{AC} &= 2 \times \vec{AB} \times \text{cis}(30°) \\ &= 2(4+2i)(\cos 30° + i\sin 30°) \\ &= 2(4+2i)(\frac{\sqrt{3}}{2} + \frac{1}{2}i) \\ &= (4+2i)(\sqrt{3}+i) \\ &= 4\sqrt{3} + 4i + 2\sqrt{3}i + 2i^2 \\ &= (4\sqrt{3}-2) + i(4+2\sqrt{3})\end{aligned}$$

Now find the point C:

$$\begin{aligned}z_C &= z_A + \vec{AC} \\ &= (-6+2i) + (4\sqrt{3} - 2 + i(4+2\sqrt{3})) \\ &= (4\sqrt{3}-8) + i(6+2\sqrt{3})\end{aligned}$$

Question 5c:
For ΔABD to be equilateral, $\vec{AD}$ is a $60°$ rotation of $\vec{AB}$ scaled by 1.

$$
\begin{aligned}
\vec{AD} &= \vec{AB} \times \text{cis}(60^\circ) \\
&= (4+2i)(\cos 60^\circ + i \sin 60^\circ) \\
&= (4+2i)(\frac{1}{2} + i\frac{\sqrt{3}}{2}) \\
&= 2 + 2i\sqrt{3} + i + i^2\sqrt{3} \\
&= (2-\sqrt{3}) + i(1+2\sqrt{3})
\end{aligned}
$$

Now find the point D:

$$
\begin{aligned}
z_D &= z_A + \vec{AD} \\
&= (-6+2i) + (2-\sqrt{3} + i(1+2\sqrt{3})) \\
&= (-4-\sqrt{3}) + i(3+2\sqrt{3})
\end{aligned}
$$

Question 6:
We analyze the quadrants of the complex numbers:

A) $-w$: Since w is in the second quadrant, $-w$ is in the fourth quadrant.

B) $2iz$: Since z is in the first quadrant, iz is in the second quadrant, and $2iz$ is also in the second quadrant.

C) $\bar{z}$: Since z is in the first quadrant, $\bar{z}$ is in the fourth quadrant.

D) $w - z$: w has a negative real part and positive imaginary part. z has a positive real part and positive imaginary part. $w - z$ will have a negative real part (negative - positive = negative) and the imaginary part depends on the relative magnitudes. However, looking at the diagram, w is 'lower' than z in terms of imaginary part or similar. Let's check: $w - z = (x_w + iy_w) - (x_z + iy_z) = (x_w - x_z) + i(y_w - y_z)$. Since $x_w < 0$ and $x_z > 0$, $x_w - x_z$ is always negative. If $y_w < y_z$, then $y_w - y_z$ is also negative. In the diagram, w is visually lower than z, so $w - z$ is in the third quadrant.

Comparing with the options, the correct answer is (D).

Question 7a:
Since $|z_1| = 1$ and $|z_2| = 1$, the points A and B lie on the unit circle. Since O is the origin, $OA = OB = 1$. In the quadrilateral $OACB$, $OA = OB$ and $AC = BC$ (by the parallelogram law of vector addition), so $OACB$ is a rhombus.

Question 7b:
The length r is the modulus of $z_1 + z_2$.

$$
\begin{aligned}
z_1 + z_2 &= (\cos\alpha + i\sin\alpha) + (\cos\beta + i\sin\beta) \\
&= (\cos\alpha + \cos\beta) + i(\sin\alpha + \sin\beta)
\end{aligned}
$$

Using the sum-to-product identities:

$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$
$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

Thus,

$$z_1 + z_2 = 2\cos\left(\frac{\alpha-\beta}{2}\right)\left[\cos\left(\frac{\alpha+\beta}{2}\right) + i\sin\left(\frac{\alpha+\beta}{2}\right)\right]$$

Since $0 < \alpha < \beta < \frac{\pi}{2}$, the term $\cos(\frac{\alpha-\beta}{2})$ is positive. Thus, the modulus r is:

$$r = 2\cos\left(\frac{\beta-\alpha}{2}\right)$$

Question 7c:
The argument θ is the angle of $z_1 + z_2$.

$$\theta = \mathrm{Arg}(z_1 + z_2) = \frac{\alpha+\beta}{2}$$

Now, using the modulus r and the argument θ, we can write the real part of $z_1 + z_2$:

$$\mathrm{Re}(z_1 + z_2) = r\cos\theta$$
$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\beta-\alpha}{2}\right)\cos\left(\frac{\alpha+\beta}{2}\right)$$

This completes the proof.